

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне
командной строки

Эргюрунлю Асуденур

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

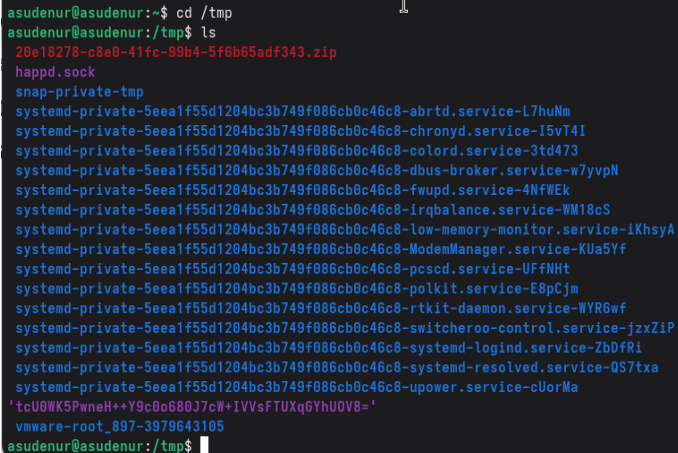
Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы

```
asudenur@asudenur:~$  
asudenur@asudenur:~$ cd  
asudenur@asudenur:~$ pwd  
/home/asudenur  
asudenur@asudenur:~$
```

Рисунок 1: Путь к домашнему каталогу



```
asudenur@asudenur:~$ cd /tmp
asudenur@asudenur:~/tmp$ ls
20e18278-c8e0-41fc-99b4-5f6b65adf343.zip
happd.sock
snap-private-tmp
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-abrtd.service-L7huNm
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-chrond.service-I5vT4I
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-colord.service-3td473
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-dbus-broker.service-w7yvpN
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-fwupd.service-4NfWEk
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-irqbalance.service-WM18cS
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-low-memory-monitor.service-iKhsyA
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-ModemManager.service-KUa5Yf
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-pcsd.service-UFfNHt
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-polkit.service-E8pCjm
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-rtkit-daemon.service-WYR6wf
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-switcheroo-control.service-jzxZiP
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-systemd-logind.service-ZbDfRi
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-systemd-resolved.service-QS7txa
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-upower.service-cUorMa
'tcUOWK5PwneH++Y9c0o680J7cW+IVVsFTUXqGYhUOV8='
vmware-root_897-3979643105
asudenur@asudenur:~/tmp$
```

Рисунок 2: Команда ls


```
asudenur@asudenur:/tmp$ ls -la
..
20e18278-c8e0-41fc-99b4-5f6b65adf343.zip
.font-unix
happd.sock
.ICE-unix
snap-private-tmp
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-abrt.service-L7huNm
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-chronyd.service-I5vT4I
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-colord.service-3td473
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-dbus-broker.service-w7yvpN
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-fwupd.service-4NfWEk
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-irqbalance.service-WM18cS
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-low-memory-monitor.service-iKhsyA
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-ModemManager.service-KUa5Yf
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-pcsd.service-UFfNHt
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-polkit.service-E8pCjm
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-rtkit-daemon.service-WYR6wf
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-switcheroo-control.service-jzxZiP
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-systemd-logind.service-ZbDfRi
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-systemd-resolved.service-QS7txa
systemd-private-5eeaf55d1204bc3b749f086cb0c46c8-upower.service-cUorMa
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXq6YhUOV8='
vmware-root_897-3979643105
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
asudenur@asudenur:/tmp$
```

Рисунок 3: Команда ls -la

```
asudenur@asudenur: /tmp$ ls -l
total 4088
-rw-r--r--. 1 asudenur asudenur 4182728 Mar 10 16:57 20e18278-c8e0-41fc-99b4-5f6b65adf343.zip
srw-rw-rw-. 1 root      root          0 Mar 10 16:57 happd.sock
drwx-----. 2 root      root          40 Mar 10 16:57 snap-private-tmp
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-abrt.service-L7huNm
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-chronyd.service-I5vT4I
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-colord.service-3td473
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-dbus-broker.service-w7yvpN
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-fwupd.service-4NfWEk
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-irqbalance.service-WM18cS
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-low-memory-monitor.service-iKhsyA
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-ModemManager.service-KUa5Yf
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-pcsd.service-UFfNHt
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-polkit.service-E8pCjm
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
6cb0c46c8-rtkit-daemon.service-WYR6wf
drwx-----. 3 root      root          60 Mar 10 16:57 systemd-private-5eea1f55d1204bc3b749f08
```

Рисунок 4: Команда ls -l

```
asudenur@asudenur:/tmp$ ls -f
.
..
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXqGYhUOV8='
20e18278-c8e0-41fc-99b4-5f6b65adf343.zip
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-fwupd.service-4NfWEk
.X1-lock
.X0-lock
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-pcsd.service-UFfNHt
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-colord.service-3td473
happd.sock
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-ModemManager.service-KUa5Yf
vmware-root_897-3979643105
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-systemd-logind.service-ZbDfRi
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-upower.service-cUorMa
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-switcheroo-control.service-jzxZiP
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-rtkit-daemon.service-WYR6wf
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-polkit.service-E8pCjm
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-low-memory-monitor.service-iKhsyA
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-chrond.service-I5vT4I
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-irqbalance.service-WM18cS
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-abrt.service-L7huNm
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-dbus-broker.service-w7yvpN
systemd-private-5eea1f55d1204bc3b749f086cb0c46c8-systemd-resolved.service-QS7txa
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
asudenur@asudenur:/tmp$
```

Рисунок 5: Команда ls -f

```
asudenur@asudenur: /tmp$  
asudenur@asudenur: /tmp$ cd /var/spool/  
asudenur@asudenur: /var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 13216 Mar 10 17:03 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 03:00 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 07:05 anacron  
drwx-----. 1 root root   18 Oct 23 07:07 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp     6 Dec 12 03:00 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail 1466 Mar  7 12:48 mail  
drwxr-xr-x. 1 root root    0 Jan 27 03:00 plymouth  
asudenur@asudenur: /var/spool$
```

Рисунок 6: Каталог /var/spool

```
asudenur@asudenur:/var/spool$ cd
asudenur@asudenur:~$ ls
Desktop    Downloads  Music      Public    snap      Videos
Documents  git-extended Pictures    site      Templates work
asudenur@asudenur:~$ ls -al
total 24
drwx-----. 1 asudenur asudenur 404 Mar  4 16:13 .
drwxr-xr-x. 1 root      root    1466 Mar  7 12:48 ..
-rw-----. 1 asudenur asudenur 5209 Mar  4 16:29 .bash_history
-rw-r--r--. 1 asudenur asudenur  18 Jul 23 2025 .bash_logout
-rw-r--r--. 1 asudenur asudenur  144 Jul 23 2025 .bash_profile
-rw-r--r--. 1 asudenur asudenur  678 Mar  4 15:55 .bashrc
drwx-----. 1 asudenur asudenur  492 Mar  4 15:56 .cache
drwx-----. 1 asudenur asudenur  364 Mar 10 17:06 .config
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Desktop
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Documents
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Downloads
drwxr-xr-x. 1 asudenur asudenur  114 Mar  4 14:48 .fontconfig
-rw-r--r--. 1 asudenur asudenur  234 Mar  4 15:49 .gitconfig
drwxr-xr-x. 1 asudenur asudenur   74 Mar  4 15:59 git-extended
drwx-----. 1 asudenur asudenur  136 Mar  4 15:48 .gnupg
drwx-----. 1 asudenur asudenur   20 Mar  4 14:39 .local
drwxr-xr-x. 1 asudenur asudenur   34 Oct 23 07:05 .mozilla
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Music
drwxr-xr-x. 1 asudenur asudenur   26 Mar  4 16:12 .npm
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Pictures
drwxr-xr-x. 1 asudenur asudenur    0 Mar  4 14:39 Public
drwxr-xr-x. 1 asudenur asudenur   50 Mar  4 16:12 site
drwx-----. 1 asudenur asudenur    8 Mar  4 16:13 snap
drwx-----. 1 asudenur asudenur  132 Mar  4 15:50 .ssh
```

Рисунок 7: Файлы в домашнем каталоге

```
asudenur@asudenur:~$  
asudenur@asudenur:~$ mkdir newdir  
asudenur@asudenur:~$ mkdir newdir/morefun  
asudenur@asudenur:~$ mkdir letters memos misk  
asudenur@asudenur:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  snap  Videos  
Documents  git-extended  memos  Music  Pictures  site  Templates  work  
asudenur@asudenur:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
asudenur@asudenur:~$ rm -r letters/ memos/ misk/  
asudenur@asudenur:~$ rm -r newdir/  
asudenur@asudenur:~$ ls  
Desktop  Downloads  Music  Public  snap  Videos  
Documents  git-extended  Pictures  site  Templates  work  
asudenur@asudenur:~$
```

Рисунок 8: Действия с каталогами

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_
extensions/yamadharma/minted-quarto':
_extension.yml  minted-quarto.lua

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/i
mage':
solvay.jpg

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_
resources':
csl  tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_
resources/csl':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_
resources/tex':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts'
:
image-report mpv-shot
asudenur@asudenur:~$ ls -t
snap  git-extended  Desktop  Downloads  Pictures  Templates
site  work            Documents  Music      Public    Videos
asudenur@asudenur:~$
```

Рисунок 9: Команда ls -R и ls -t

```
BASH_BUILTINS(1)                                General Commands Manual                                BASH_BUILTINS(1)

NAME
: , . , [ , alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt,
continue, declare, dirs, disown, echo, enable, eval, exec, exit, export, false, fc, fg,
getopts, hash, help, history, jobs, kill, let, local, logout, mapfile, popd, printf,
pushd, pwd, read, readarray, readonly, return, set, shift, shopt, source, suspend, test,
times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in com-
mands, see bash(1)

BASH BUILTIN COMMANDS
Unless otherwise noted, each builtin command documented in this section as accepting op-
tions preceded by - accepts -- to signify the end of the options. The :, true, false, and
test/[ builtins do not accept options and do not treat -- specially. The exit, logout,
return, break, continue, let, and shift builtins accept and process arguments beginning
with - without requiring --. Other builtins that accept arguments but are not specified
as accepting options interpret arguments beginning with - as invalid options and require
-- to prevent this interpretation.

: [arguments]
    No effect; the command does nothing beyond expanding arguments and performing any
    specified redirections. The return status is zero.

. [-p path] filename [arguments]
source [-p path] filename [arguments]
    The . command (source) reads and execute commands from filename in the current
    shell environment and returns the exit status of the last command executed from
    filename.

Manual page cd(1) line 1 (press h for help or q to quit)
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described
    here. Please refer to your shell's documentation for details about the options it sup-
    ports.

Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes un-
        affected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to
        CTX

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
        a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

  If the -i or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
    ignore nonexistent files and arguments, never prompt

  -i      prompt before every removal

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
276 cd /tmp
277 ls
278 ls -a
279 ls -l
280 ls -f
281 cd /var/spool/
282 ls -l
283 cd
284 ls
285 ls -al
286 mkdir newdir
287 mkdir newdir/morefun
288 mkdir letters memos misk
289 ls
290 rm letters/ memos/ misk/
291 rm -r letters/ memos/ misk/
292 rm -r newdir/
293 ls
294 ls -R
295 ls -t
296 help cd
297 man pwd
298 man mkdir
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.